

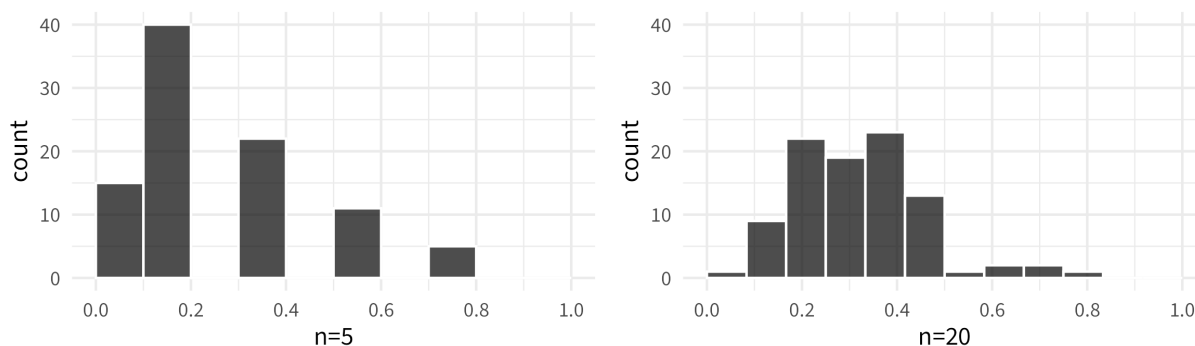
# NOTES 11: CONFIDENCE INTERVALS + INTRO TO THE BOOTSTRAP

Stat 120 | Spring 2026  
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“Big picture” picture

**Sampling distributions** for fingerprint example from last time (random samples)



As \_\_\_\_\_ increases:

The mean and standard deviation of the sampling distribution are:

```
# A tibble: 2 x 3
  sample mean    sd
  <chr>   <dbl> <dbl>
1 n=20    0.332 0.132
2 n=5     0.298 0.210
```

## Standard Error:

Standard deviation of a statistic or a sampling distribution

What's the **point estimate** for  $p_{\text{whorl}}$  using the  $n = 20$  sample? The  $n = 5$ ?

Would you be surprised if  $p_{\text{whorl}}$  was actually 0.39? What about 0.80?

### Confidence Interval

An interval computed from sample data by a method that will capture the parameter for a specified percentage of all samples

95% confidence interval using the standard error:

### Margin of Error

Range of values above and below a point estimate that defines the boundary of a confidence interval

#### CI Misinterpretations

- A 95% CI contains 95% of the data from the population
- I am 95% sure that the mean of the sample will be in CI
- The probability that the parameter is in the CI is 95%

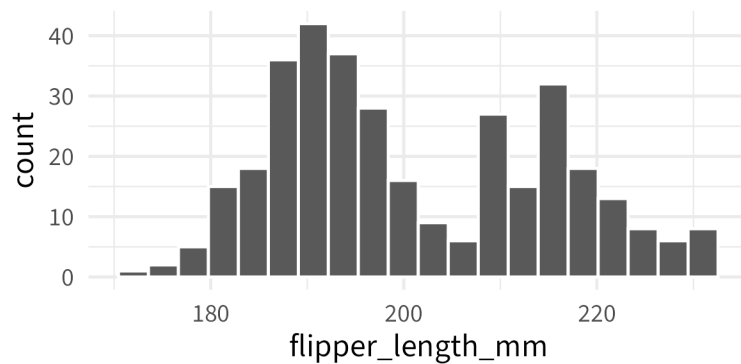
Structure of a CI:

1. I am \_\_\_\_ % confident
2. that the [population parameter in context]
3. is between \_\_\_\_ and \_\_\_\_ [units]

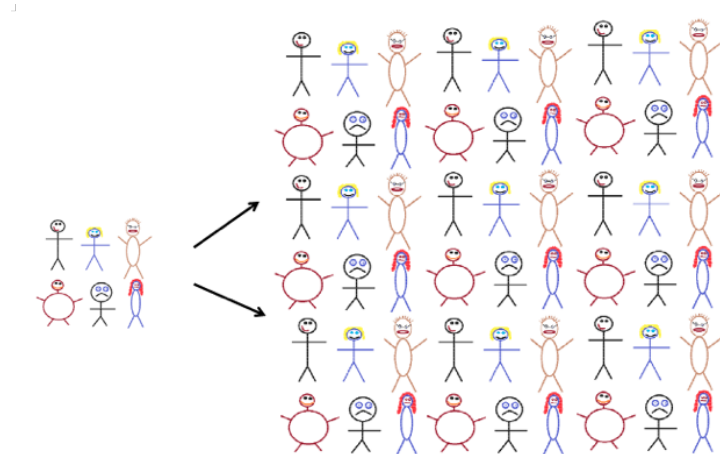
# 1 The Bootstrap

So far, we've relied on generating lots of samples to construct sampling distributions (because we relied on the fingerprint sampler). For most situations, this is unrealistic.

**Example:** Let's consider our penguin friends. I'm interested in finding a 95% confidence interval for the mean flipper length of *all* Palmer Archipelago penguins. The mean flipper length is 200.9152 and the standard deviation is 14.0617.

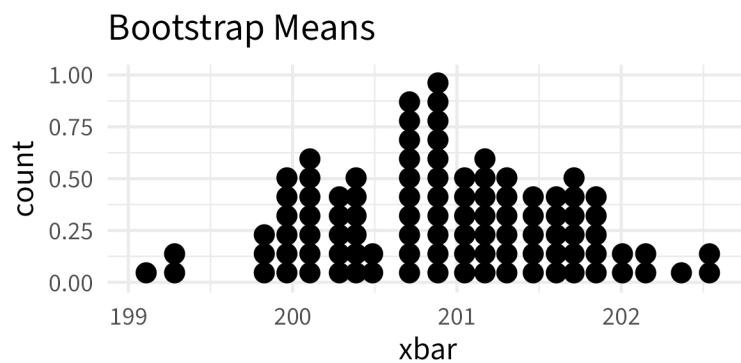


**Idea:** if we make a bunch of copies of our random sample, we should get something close to the population



**Bootstrap picture:**

Let's return to our penguins. I've made 100 bootstrap samples and found the mean for each one:



The mean of this distribution is 200.9209 and the standard deviation is 0.7329.

#### Bootstrap confidence interval:

The magic of the bootstrap is that it works for any statistic we can compute from our sample! We simply have to follow these steps:

1. Generate *bootstrap samples*:
  - Sample from the original sample with replacement
  - Use the same sample size as original sample
2. Compute the statistic of interest for each bootstrap sample
3. Collect the statistics for many bootstrap samples to create the *bootstrap distribution*
4. Treat the *bootstrap distribution* as the *sampling distribution* to estimate the standard error

⚠ What happens when the bootstrap distribution is not symmetric?